

BRAC UNIVERSITY
Department of Computer Science and Engineering

Quiz 02
Semester: Fall 2025

Duration: 20 min
Full Marks: 15

Name:	ID:	Section:
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CSE 260: Digital Logic Design
SET- A

1. [CO2] Consider the following boolean equation and answer the following questions, $F(X,Y,Z) = XY'Z + Y'Z' + (XY)'Z$
 - a. Simplify the boolean expression using boolean algebra (5 Marks)
 - b. Draw the simplified expression using NOR gate only (3 Marks)
2. [CO2] (6 + 1 Marks) Convert the following equation to its canonical SOP form and find the maxterms, $F(a,b,c,d) = ab' + bc + a'c'$
Note: You can not simplify this equation